

Network Scanner Scripts

Pulling, running, and configuring `network_scanner.py` and `mobile_network_scanner.py`

Repository: <https://github.com/thomasmd321/Test-repo>

Branch: `claude/local-network-device-discovery-joawq5`

Both scripts discover devices on your local network with no external service and no account required. **network_scanner.py** targets a normal desktop/Termux environment and can read ARP/MAC data directly; **mobile_network_scanner.py** targets a sandboxed environment such as an iPhone running a-Shell, where the OS blocks raw sockets and subprocess calls, so it relies on plain TCP connections instead.

	<code>network_scanner.py</code>	<code>mobile_network_scanner.py</code>
Target platform	Linux / macOS / Windows / Termux	iOS sandboxes (a-Shell, Pythonista)
Discovery method	ARP scan (scapy), ping-sweep fallback	Plain TCP connect probe
MAC address / vendor	Yes (IEEE OUI lookup)	No (not exposed to the sandbox)
IPv6 discovery	Yes (<code>--ipv6</code>)	No
Banner grabbing	<code>--identify IP deep dive</code>	Built into every scan
Risky-port flagging	Yes	Yes
NEW / CHG / missing tracking	Yes	Yes
Export results	<code>--output FILE (CSV/JSON)</code>	<code>--output FILE (CSV/JSON)</code>
Custom device labels	<code>--set-label / --remove-label</code>	<code>--set-label / --remove-label</code>
Environment diagnostics	<code>--doctor</code>	<code>--doctor</code>
Quiet mode (cron/systemd)	<code>--quiet</code>	<code>--quiet</code>

1. Getting the scripts

Both scripts are plain, dependency-light Python files living in one repository. Pick whichever pull method fits where you're running them.

Clone the whole repository (desktop / Termux)

```
git clone https://github.com/thomasmd321/Test-repo.git
cd Test-repo
git checkout claude/local-network-device-discovery-joawq5
```

Grab a single file directly (no git needed)

Useful on iOS, or anywhere you only want one script without the rest of the repo:

```
BASE=https://raw.githubusercontent.com/thomasmd321/Test-repo/claude/local-network-device-discovery-joawq5
curl -O $BASE/network_scanner.py
curl -O $BASE/mobile_network_scanner.py
```

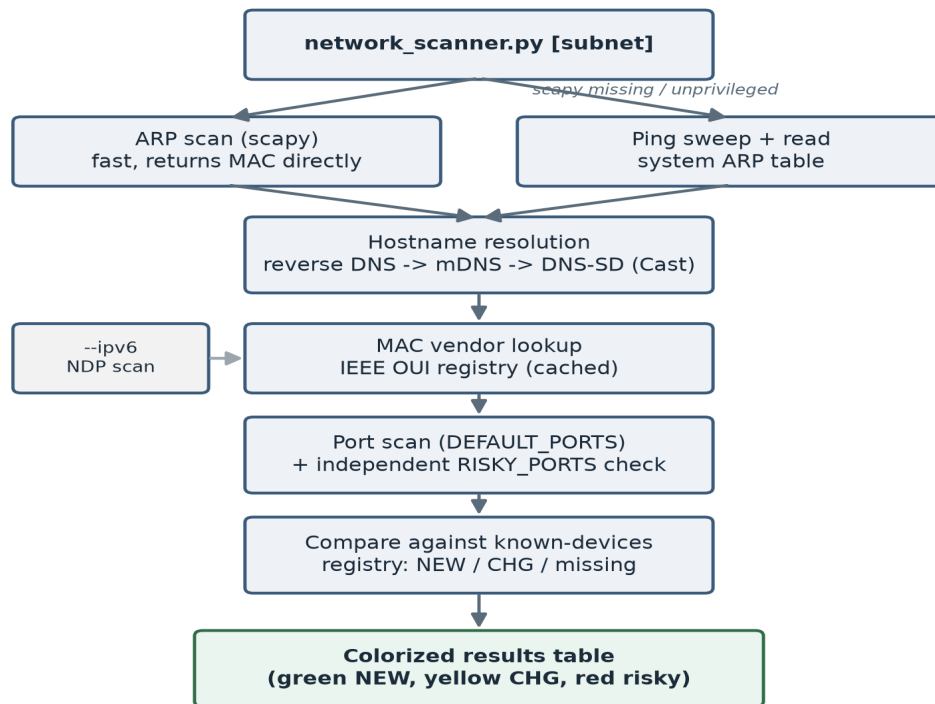
`curl -O` saves the file under its own name in the current directory and overwrites an existing copy, so re-running the same command later pulls in any updates.

On an iPhone

Install **a-Shell** from the App Store (not “a-Shell mini,” which strips out `git`), then run either the clone or the single-file `curl` command above from its terminal.

2. network_scanner.py

For a normal desktop, laptop, or Termux-on-Android environment. Prefers a fast ARP scan via scapy (returns MAC addresses directly); if scapy isn't installed or the process lacks the privileges ARP scanning needs, it falls back automatically to a multithreaded ping sweep plus the system's own ARP table — no flag needed to trigger this, it just happens.



Falls back automatically when scapy / raw sockets aren't available.

How a scan flows through network_scanner.py, start to finish.

Running it

```
python3 network_scanner.py                # auto-detect subnet
python3 network_scanner.py 192.168.1.0/24  # scan a specific subnet
python3 network_scanner.py --all-subnets   # every subnet (needs psutil)
python3 network_scanner.py --identify 192.168.1.42 # deep-dive one host
python3 network_scanner.py --watch 300      # rescan every 5 minutes
python3 network_scanner.py --output scan.json # save results to a file
```

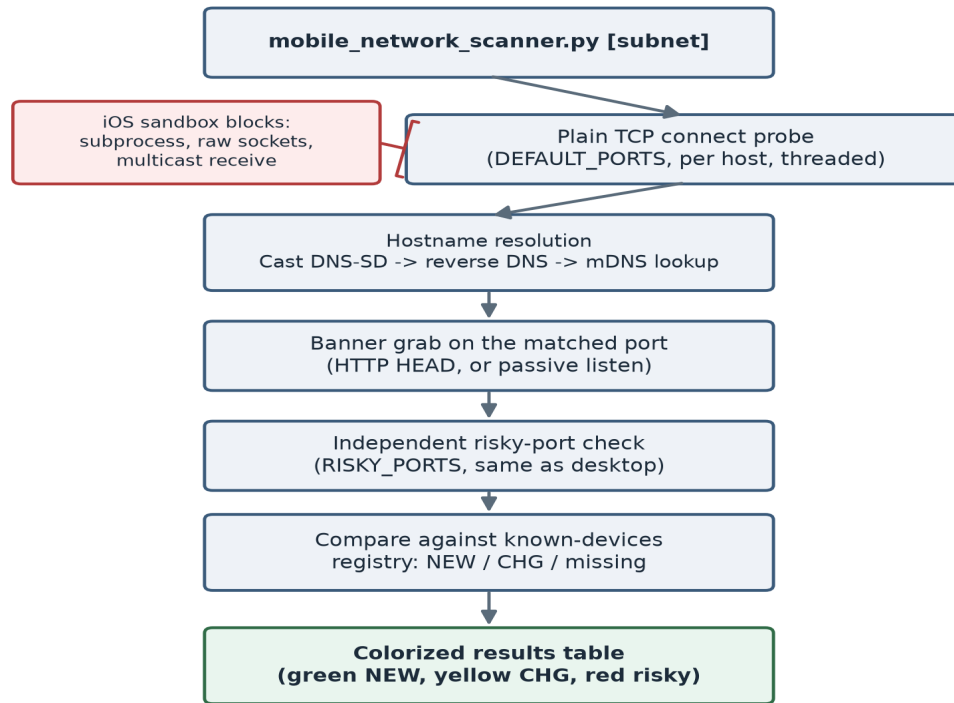
Options

Flag	What it does
subnet	Positional. CIDR range(s) to scan, comma-separated for more than one. Auto-detected if omitted.
--all-subnets	Scan every local subnet this machine has an interface on (requires <code>pip install psutil</code>) instead of just the default route.

Flag	What it does
--identify IP	Skip the network scan; do a slow, thorough single-host investigation instead (more ports, banner grabs, full hostname/vendor resolution).
--doctor	Skip the network scan; check this environment for everything this script can use (scapy, ping/arp, cache writability, mDNS, etc.) and report it.
--timeout SECONDS	Timeout per host (default: 1.0).
--mdns-timeout SECONDS	Timeout for the mDNS/DNS-SD hostname fallback (default: 0.3).
--no-vendor-lookup	Skip the IEEE OUI vendor lookup (avoids the first-run registry download).
--refresh-vendor-db	Force a fresh OUI registry download instead of using the cached copy.
--watch SECONDS	Rescan on a timer instead of once, printing NEW markers as they appear (Ctrl+C to stop).
--no-track-devices	Don't use the known-devices registry at all — no NEW/CHG markers, nothing remembered.
--forget-known-devices	Clear the registry first, so everything in this run shows as NEW.
--set-label KEY=LABEL	Assign a friendly label to a device (KEY is its MAC, or IP if it has none), shown instead of/alongside its hostname. Repeatable.
--remove-label KEY	Remove a device's custom label. Repeatable.
--ipv6	Also discover IPv6 devices via multicast ping + NDP (Linux/macOS only).
--ipv6-timeout SECONDS	Roughly how long to spend on IPv6 discovery (default: 2.0).
--no-scan-ports	Skip the open-port probe (and the risky-ports check that depends on it).
--ports LIST	Comma-separated TCP ports to probe instead of the built-in default list.
--port-timeout SECONDS	Per-port connection timeout, for both the port probe and risky-ports check (default: 0.3).
--no-risky-ports	Skip the risky-ports security check while keeping the general port probe.
--no-color	Disable ANSI color output (also respects the NO_COLOR env var).
--output FILE	Save this scan's results to FILE as JSON, or CSV if it ends in .csv.
--quiet	Print nothing for a scan with no NEW/CHG/missing/risky devices - only an interesting run produces output (see --watch under cron/systemd).

3. mobile_network_scanner.py

For sandboxed Python runtimes — most notably iOS apps like a-Shell — that allow neither subprocess calls nor raw sockets. Every discovery technique here is built from ordinary client TCP/UDP sockets, which is the one thing every one of these sandboxes still permits.



No ARP, no IPv6 discovery, no MAC vendor lookup - ordinary sockets only.

How a scan flows through mobile_network_scanner.py, start to finish.

Running it

```
python3 mobile_network_scanner.py          # auto-detect subnet
python3 mobile_network_scanner.py 192.168.1.0/24 # scan a specific subnet
python3 mobile_network_scanner.py --timeout 0.5 --ports 22,80,443
python3 mobile_network_scanner.py --watch 300      # rescan every 5 minutes
python3 mobile_network_scanner.py --no-banners --no-risky-ports # faster scan
python3 mobile_network_scanner.py --output scan.csv # save results to a file
```

Options

Flag	What it does
subnet	Positional. CIDR range(s) to scan, comma-separated for more than one. Auto-detected if omitted.
--doctor	Skip the network scan; check this environment for everything this script can use (subnet detection, TCP connectivity, cache writability, mDNS) and report it.
--timeout SECONDS	Timeout per port probe (default: 0.5); also used for the banner-grab and risky-ports steps.

Flag	What it does
--ports LIST	Comma-separated TCP ports to probe instead of the built-in default list.
--mdns-timeout SECONDS	Timeout for the mDNS/Bonjour hostname fallback (default: 0.3).
--watch SECONDS	Rescan on a timer instead of once, printing NEW markers as they appear (Ctrl+C to stop).
--no-track-devices	Don't use the known-devices registry at all — no NEW/CHG markers, nothing remembered.
--forget-known-devices	Clear the registry first, so everything in this run shows as NEW.
--set-label KEY=LABEL	Assign a friendly label to a device (KEY is its IP address), shown instead of/alongside its hostname. Repeatable.
--remove-label KEY	Remove a device's custom label. Repeatable.
--no-banners	Skip banner grabbing on each device's open port — faster, but a weaker identification hint.
--no-risky-ports	Skip the risky-ports security check while keeping the general port probe.
--no-color	Disable ANSI color output (also respects the NO_COLOR env var).
--output FILE	Save this scan's results to FILE as JSON, or CSV if it ends in .csv.
--quiet	Print nothing for a scan with no NEW/CHG/missing/risky devices - only an interesting run produces output (see --watch under cron/systemd).

iOS limitation: mDNS/DNS-SD queries fail outright on iOS with `OSError(65, 'No route to host')` — Apple's Local Network Privacy model requires an app bundle to declare Bonjour service types in advance, which a script run from a generic terminal app can't do. Plain TCP port scanning, banner grabbing, and risky-port checks are unaffected, since they're ordinary unicast traffic.

4. Known-device tracking, color coding, and --watch

Both scripts persist a small local registry of every device they've ever seen

(`~/.cache/network_scanner_known_devices.json` and

`~/.cache/mobile_network_scanner_known_devices.json`). Every scan is compared against it, producing the markers and colors below. Pair either script with `- -watch SECONDS` to turn a one-shot scan into a standing monitor.

```
python3 network_scanner.py

Scanning 192.168.1.0/24 ...

  IP Address      MAC Address      Vendor      Port  Service  Hostname
-----
NEW 192.168.1.42   3c:22:fb:1a:9c:04 Apple, Inc  62078 lockdownd iPhone.local
192.168.1.1      aa:bb:cc:dd:ee:ff Netgear    80    http    router.local
CHG 192.168.1.57   10:2c:6b:44:2a:11 Sonos      23    ?       kitchen-speaker.local
192.168.1.72     b0:c5:54:2f:88:19 Google     8009   chromecast Living Room TV
192.168.1.88     f4:5e:ab:12:34:56 Synology   445    smb     nas.local      (risky)

5 device(s) found. 1 new since last seen.
2 previously-seen device(s) not found in this scan:
192.168.1.63      last seen 2026-09-14T08:03:00 (laptop.local)
1 device(s) with a changed port since last seen:
192.168.1.57      (none) -> telnet (23)
▲ 1 device(s) exposing commonly-risky ports:
192.168.1.88     f4:5e:ab:12:34:56 smb (445)
```

Illustrative output — exact columns vary slightly between the two scripts.

Marker / color	Meaning
NEW (green)	First time this device has ever been seen.
CHG (yellow)	A known device is answering on a different port than last time.
red row	Device exposes a port on the risky-ports list (telnet, SMB, RDP, VNC, FTP).
dim gray line	A previously-seen device that didn't show up in this scan.

A device can be NEW or CHG, never both — a device with no prior registry entry can't have a “changed” port, only a first one. Risky-row coloring takes priority over both if a device qualifies for more than one.

Resetting tracking

```
python3 network_scanner.py --no-track-devices      # skip tracking this run
python3 network_scanner.py --forget-known-devices  # reset registry, all NEW
python3 mobile_network_scanner.py --watch 300      # standing monitor on phone
```

Custom device labels

The same registry can hold a friendly label for a device — useful when its real hostname is cryptic or blank — shown in place of/alongside the hostname (e.g. Kitchen Server (localhost) when they differ). KEY is a device's MAC (or IP if it has none) for network_scanner.py, or always its IP for mobile_network_scanner.py.

```
python3 network_scanner.py --set-label aa:bb:cc:dd:ee:ff="Kitchen Server"
python3 mobile_network_scanner.py --set-label 192.168.1.42="Kitchen Echo"
python3 network_scanner.py --remove-label aa:bb:cc:dd:ee:ff
```

Both flags are repeatable and take effect immediately, including on the scan that runs in the same command. A device doesn't need to already be in the registry — labeling one creates a minimal entry for it, though that also means it won't show as NEW the next time it's actually scanned.

5. Exporting results

Both scripts can save a scan's results to a file with `--output FILE`, independent of the known-devices registry above — useful for feeding results into another tool, diffing two scans, or just keeping a dated record. The format is chosen by the file extension: `.csv` writes a CSV file, anything else (typically `.json`) writes JSON.

```
python3 network_scanner.py --output scan.json
python3 network_scanner.py --output scan.csv
python3 mobile_network_scanner.py --output scan.json
```

Each run overwrites FILE with that scan's results — this is a snapshot of the latest scan, not an appended history log.

6. Diffing two scans

scan_diff.py is a third, standalone script that compares two files saved with --output above and reports what changed: devices added, devices that disappeared, and per-field changes (a different port, hostname, banner, etc.) on devices present in both — the same NEW/missing/CHG comparison the scanners do against their own known-devices registry, just applied to two snapshots on disk instead.

```
python3 scan_diff.py old.json new.json
python3 scan_diff.py before.csv after.csv
python3 scan_diff.py --no-color old.json new.json
```

Either file can be JSON or CSV independently (comparing a CSV export against a JSON one works fine), and they don't need to come from the same script: fields the two scanners don't share (mac/vendor are desktop-only, banner is mobile-only) are still compared whenever both files happen to have them. A device is matched across the two files by MAC when present, falling back to IP — the same identity rule network_scanner.py's own known-devices tracking uses.

7. Quiet mode and environment diagnostics

--quiet

Suppresses everything — even the scan's own “Scanning...” line — for a run with nothing to report: no NEW devices, no port changes, no missing devices, and no risky ports. As soon as one of those is true, the full report prints exactly as it would without --quiet. Meant for --watch under cron/systemd, where a boring rescan producing zero output (rather than a full table every time) is what makes “did anything happen” easy to grep for or alert on.

```
python3 network_scanner.py --watch 300 --quiet
python3 mobile_network_scanner.py --watch 300 --quiet
```

--doctor

Skips the network scan and instead checks this environment for everything the script can use — scapy, ping/arp, cache-directory writability, mDNS multicast, and so on — reporting each with a clear pass/fail and, for most, the fallback that kicks in when it fails. Exits 0 if every check passed, 1 otherwise, so it's usable as a pre-flight check in a script.

```
python3 network_scanner.py --doctor
python3 mobile_network_scanner.py --doctor
```

A failed check doesn't necessarily mean a scan will fail — most have a documented fallback (no arp on PATH just means no MAC from that path, not a crash). The one exception: a failed mDNS check on `mobile_network_scanner.py` almost always means iOS's Local Network Privacy restriction (see `mdns_diagnostic.py`), which no retry or code change here can fix.